

Void Phase Dynamics: The Physical Principles of Classical Philosophy

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Abstract

This paper proposes a unified physical framework conjecture named "Void Phase Dynamics" (VPD). The theory takes "void fluctuation" as the sole underlying form of motion in the universe, and argues that spacetime is not the stage on which quantum fluctuations depend for their existence, but rather the product of quantum fluctuations themselves. The core of the theory includes: the alternating-universe model and the natural solution of absolute zero, the cosmological constant explanation with zero vacuum energy, the phase-consistency explanation of proton longevity, the explanation of gravitational mass and inertial mass as mutual action and reaction, the symmetrized Maxwell equations and the magnetic monopole explanation, the explanation of dark matter and dark energy through three-phase time and six-direction space parallel universes, as well as a unified description of wave-particle duality, quantum field theory compatibility, nuclear energy mechanisms, and the hierarchical structure of the universe. VPD does not overturn existing theories, but only slightly modifies them; existing formulas remain applicable. This paper aims to provide an underlying physical framework that can be discussed, tested, and deduced.

Keywords: Void Phase Dynamics; void fluctuation; alternating universe; absolute zero; vacuum energy; proton longevity; magnetic monopole; dark matter; dark energy; parallel universes; wave-particle duality; quantum field theory; hierarchical structure of the universe

1. Introduction

Physics has progressed through the absolute spacetime view of Galileo and Newton, the cancellation of absolute time by special relativity, the cancellation of the special status of global inertial frames by general relativity, and the cancellation of the privilege of "absolute vacuum" by quantum field theory. Each advance has been the cancellation of some kind of "absolute background." Extending this logic to its ultimate conclusion: the spacetime background itself is no longer the stage on which quantum fluctuations depend for their existence, but rather the product of quantum fluctuations themselves. From "fluctuations in spacetime" to "fluctuations generating spacetime," the ultimate cancellation of the absolute status of spacetime is achieved. This is the core conjecture of Void Phase Dynamics.

This theory attempts to solve the following problems:

1. The singularity energy problem; absolute zero is the natural solution (alternating universe model).
2. The cosmological constant problem; vacuum energy is 0.
3. The proton longevity problem (whether something is a proton depends on whether it is phase-consistent with the emergent subject).
4. Gravitational mass and inertial mass (mutual action and reaction).

5. The magnetic monopole problem (modifying Maxwell's equations to be symmetric).
 6. The existence problem of dark matter and dark energy (three-phase time and six-direction space parallel universes).
 7. Unifying the underlying form of motion as void fluctuation, without need for linear motion or rotation.
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2. Theoretical Framework

2.1 Void Fluctuation: The Underlying Form of Motion

Void fluctuation is divided into "rise" and "fall," with a periodic sequence of {0, +1, 0, -1}, and a period precision of 3 bits, totaling 8 phase levels. The integral of the direction of fluctuation motion manifests as electric charge: the integral of rise (ascent) manifests as positive charge, and the integral of fall (descent) manifests as negative charge; the integral of fluctuation manifests as charge, the source of fractional charge. Electric current is the integral of the action of charge on the macroscopic scale. Magnetism is the manifestation of changes between charges existing simultaneously at different points in space, with electric field lines closed, curl, the magnetic field plane perpendicular to the line connecting charges, magnetic poles simultaneously observable, and relative to the observer as a third party. Light is the manifestation of charge changes at the same point in space at different moments in time, with electric field lines extending to infinity, divergence, the magnetic field plane perpendicular to the observer, magnetic monopoles, light, 2 types of spin corresponding to N and S poles, and relative to the observer as a second party.

Fermions: a pair of interactions existing simultaneously in space, with each of the 2 points undergoing one fluctuation within one period, totaling 2 cycles, and spin manifesting as $1/2$. Bosons: a pair of interactions existing successively in time, with only one fluctuation experienced within one period, only 1 cycle at a single point, and spin manifesting as 1.

An atom is an oscillation sequence on the hollow space formed by fluctuations. The central oscillation period does not manifest an effect on the atom; it is merely the pole where the oscillation direction changes. Electrons do not exist; the electron cloud is the residual image left by the overall oscillation of the atom.

2.2 Spacetime Generation and Matter Annihilation-Rebirth

Vacuum fluctuation generates spacetime; spacetime is a kind of motion, and is matter itself. To discuss matter apart from motion is for matter not to exist; to discuss spacetime apart from matter is for spacetime not to exist. After matter is born, it belongs to that spacetime point; it cannot move or rotate, and can only annihilate in place. Spacetime can be imagined as an empty shell, with the shell layer being matter and the interior being spacetime. When the shell layer manifests as positive matter, spacetime is a negative-energy space; when the shell layer manifests as antimatter, spacetime is a positive-energy space. In topology, this corresponds to the inversion of the inside and outside of a spherical surface.

After fluctuation generates spacetime, spacetime transmits the information of its birth to all neighbors within the range of the force. After the spacetime annihilates, all neighbors jointly determine the rebirth of spacetime at that spacetime point. This reaction manifests as a field. The pole after rebirth manifests as a particle, and the process of annihilation and rebirth manifests as a wave.

2.3 Absolute Zero and the Alternating Universe

Absolute zero corresponds to the zero-crossing moment of the change between positive and anti-matter (the natural solution of this theory). The CMB corresponds to the instantaneous power value of matter in phase. The infinite energy base of the Big Bang singularity is cancelled; the universe is born from a void breach, a veritable cold explosion (hot explosion $\rightarrow$ cold explosion). The original noisy pulsating DC universe

is replaced by a noisy AC universe. The cost is that matter does not persist; we are instantaneous emergence. This cost is better than our being virtualized by another advanced civilization, and it also does not require divine creation. Of course, one could also hold that there exists a cell-like renewal similar to the human body, that is, the underlying layer is instantaneously annihilated and reborn, while the upper layer exists permanently, but this requires adding more conditions and does not conform to the first principle. This theory is more inclined to the view that we are instantaneously generated, which provides a more concise foundation for deriving classical philosophy later.

2.4 The Mystery of Proton Immortality

The reason why proton lifetime differs from that of other fundamental particles is that standard theory assumes baryon number conservation, the lightest baryon, and a pile of other assumptions. Except for photons and electrons, all other particles are short-lived. The stability of neutrons in nuclei forms a contrast with the decay of free neutrons. This theory explains: the particle that emerges as us observers is the proton. When the proton annihilates, the observer annihilates, so the annihilation of the proton is not observed by itself. This is the same as a person not being able to see their own death. Definition of proton: a proton is a particle whose phase is consistent with the emergent observer subject; all others are not protons. The phase difference from the observer determines the mass manifestation of the particle. Neutrons in nuclei are affected by the self-sustaining emergence of the atomic subject, and their phase is constrained within the error range. The phase of free neutrons, through multiple accumulations of error, when reaching the threshold with the emergent subject, slides toward the antimatter space and undergoes energy settlement to release photons (determined by the precision of the phase difference).

2.5 Motion and Field, Wave-Particle Duality

All nouns in human language will correspond to a verb. Vacuum fluctuation generates spacetime; spacetime is a kind of motion, and is matter itself. After matter is born, it belongs to that spacetime point; it cannot move or rotate, and can only annihilate in place. After fluctuation generates spacetime, spacetime transmits the information of its birth to all neighbors within the range of the force. After the spacetime annihilates, all neighbors jointly determine the rebirth of spacetime at that spacetime point. This reaction manifests as a field. The pole after rebirth manifests as a particle, and the process of annihilation and rebirth manifests as a wave.

2.6 Fluctuation Phase and the Origin of Mass

The phase precision is 3 bits, totaling 8 levels. The atom is the smallest particle constituting structure, and there is no meaning in further dividing it. A proton is an atom consistent with the emergent subject. A neutron is an atom with a relatively large phase difference from the emergent subject. Protons and neutrons are images of phase; quarks are images of the atom viewed from within. The spin of the atom is the regular cycle of annihilation and rebirth (the running-light metaphor). The atom is the pole at the center where the direction of the motion phase changes; there is no atomic nucleus. All particles are images of motion, not real particles. The changes of particles are different manifestations of the change in the phase difference from the emergent subject. Mass is not an attribute of particles; it is the phase delay caused by the collective queuing and sliding rebirth of particles. This is caused by the characteristics of fermions, and does not require the Higgs boson to bring mass.

2.7 Nuclear Energy Mechanism

What is distributed at different phase intervals in space is positive and negative fluctuation motion. These fluctuations all perform uniform oscillation according to the laws of motion (the state of motion is maintained unchanged when not subject to external forces). These fluctuations proceed in an orderly manner at their respective spacetime points, and the event of positive and negative matter meeting each

other and undergoing violent annihilation will not occur. When destroyed by an external force, violent changes in the phase of motion will occur; mutually adjacent positive and negative motions will become chaotic, and chain-like energy settlements will continuously occur, releasing large numbers of photons (like a chain rear-end collision on a highway), except that this is a motion-phase rear-end collision. Note that there is no linear motion and no rotation, only in-place fluctuation, so rear-end collisions are all violent changes of phase in place. This violent change will continuously release photons; generating photons is generating new spacetime fluctuations. There is only motion, no particles; particles are all names adapted from the past. That is, the original regular emergence becomes irregular emergence, similar to the early Big Bang.

2.8 Compatibility with General Relativity

The substrate of force action described by general relativity is not the underlying layer of spacetime, but corresponds to spacetime structure. The quantized empty substrate in the central region of the universe is flat; what is curved is the force-transmitting structure. This structure transmitting force is easy to understand; structures such as electric wires and CPUs are good proof. The edge of the universe has always been growing at the speed of light, a chaos, suitable for GR description. In the central region of the universe, regional centers of mass generally exist. Particles slide toward the center of mass each time under the action of neighbors; for small-range centers of mass, this is quickly averaged out. The emergent huge positive and negative matter centers of mass form an event horizon on the midline, and positive and negative matter slide toward each other respectively. Note that this regional center-of-mass sliding does not cause matter concentration, but causes matter annihilation: it is positive and negative matter sliding toward each other respectively, not sliding toward matter of the same phase as themselves. This is a kind of death-seeking, maintaining the universe's energy breach at zero.

2.9 Cosmological Explanation

Black holes, large-scale antimatter aggregation regions, form Lagrange surfaces; positive matter slides toward the antimatter region, and antimatter slides toward the positive matter region. Positive matter slides toward the antimatter region, and antimatter slides toward the positive matter region. The result is mutual annihilation, forming large patches of void regions, the source of the event horizon. Hubble tension redshift, similarly to black holes, also produces similar sliding effects in tiny black hole regions. The formation of large-scale structure: the universe is a structure based on growth, which is the foundation for forming large structures. Positive matter aggregation regions, antimatter aggregation regions, and large-scale average matter regions (cosmic voids) will form. Big Bang → cold explosion; the inflation mechanism corresponds to the early chaotic period of the universe, before the resonance period. Baryon acoustic oscillations (BAO): resonance of the spacetime lattice may produce specific scales. Time dilation: objects moving at high speed skip spacetime lattice points upon rebirth; phase synchronization leads to fewer oscillation counts. The ultimate fate of the universe: forever growing at the speed of light.

2.10 Compatibility with Quantum Field Theory

Wave function: this rebirth broadcasts its own information to neighbors, and after annihilation it is reborn under the collective action of neighbors. This collectively determined probability is the wave function. Path integral: the particle does not traverse all paths, but annihilates in place and is then reborn at the next point under the action of neighbors. Mathematically, this corresponds to the action of the wave function. Quantum tunneling: this is simply that after a particle annihilates, another particle passes through this annihilation gap. Quantum entanglement: the manifestation of two particles with mutually aligned phases maintaining their original motion unchanged when not subject to external forces; the phase always remains unchanged relative to the overall phase, and remains unchanged relative to the phase difference of the emergent observer. Quantum decoherence: the phase alignment between observer and observed, the

overall emergence of the spacetime lattice phase, the phase difference unchanged, the observation result stable, the emergence speed extremely fast, giving the feeling of continuity.

2.11 Hierarchical Structure of the Universe

A layered cosmic structure, with each layer running an algorithm similar to the forward/backward propagation of artificial intelligence. Void fluctuation → atom → structure → consciousness. Each layer forms a self-sustaining structure, and at the same time performs reverse transmission (gravity) to the underlying layer. This process is very similar to current artificial intelligence generation algorithms, with forward transmission (emergence) and reverse transmission (gravity). We are a huge self-generating computer.

2.12 Qi Cultivation Principle

Molecular structure → consciousness → consciousness reverse → molecular reverse → atomic reverse → fluctuation spacetime → atomic structure → molecular structure. Qi cultivation: starting from this molecular structure → a cycle that affects itself again → molecular structure.

2.13 Three-Phase Time

$$[y(\theta) = \frac{\int_0^\theta \sin(2t)^p dt}{\int_0^{\pi/2} \sin(2t)^p dt}]$$

Oscillation period: {0, +1, 0, -1, 0}. Positive matter, antimatter, emergence three phases (corresponding to the yang, ghost, and yin realms of classical philosophy).

2.14 Space 6 Directions × Time 3 Phases: 18 Parallel Universes in Total

Because there is a time difference (phase difference) between the moment of matter rebirth and the occurrence of interaction between observer and observed, matter and dark matter must be less than the theoretical value, and the positive error is all given to dark energy. Observed values: matter 4.9%, -0.65%; dark matter 26.8%, -0.95%; dark energy 68.3%, +1.7%. Total dark matter phase difference: posterior phase difference. Spacetime reuse technology, similar in principle to time-division multiple access and frequency-division multiple access in communications.

2.15 Symmetrization of Maxwell's Equations

Asymmetry = ugliness; symmetry = magnetic monopole (photon).

$$[\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}]$$

$$[\nabla \cdot \mathbf{B} = 0]$$

$$[\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}]$$

$$[\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}]$$

$$[\mathbf{F} = \mathbf{E} + ic\mathbf{B}]$$

$$[\nabla \cdot \mathbf{F} = \frac{\rho}{\epsilon_0}]$$

$$[\nabla \times \mathbf{F} = ic\mu_0 \mathbf{J} + \frac{i}{c} \frac{\partial \mathbf{F}}{\partial t}]$$

Divergence has no (i), corresponding to time change; curl has (i), corresponding to spatial closure.

2.16 Vacuum Energy and the Cosmological Constant

Vacuum energy is zero. Probability amplitude and probability: probability amplitude $\psi = \sin\theta$ has positive and negative values (phase), probability $|\psi|^2 = \sin^2\theta$ is always non-negative, radius $r = r_{\max} \sin\theta$ has positive and negative values (direction), potential energy $E_p \propto \sin^2\theta$ is always non-negative. Standard quantum field theory: $\langle 0|H|0\rangle = \frac{1}{2}\hbar\omega \neq 0$, leading to the cosmological constant problem. In this framework, vacuum energy is zero, naturally solving this problem.

2.17 The Deadlock of Modern Physics

We have always believed that we exist continuously. In fact, we do not exist continuously; we are only instantaneous emergence, occupying only 1/3 of the time period and only 1/6 of the spatial position, which combined means we occupy only 1/18, and the rest is dark matter and dark energy. The other 5 portions in space that exist simultaneously with us are dark matter, and the 12 portions that do not exist simultaneously with us are dark energy. Please try to imagine what it would be like if we did not exist continuously, and then deduce all experimental results mathematically once again; all problems will suddenly become clear. All experiments, including personal observation, instrument observation, and particle colliders, depend on this instantaneous emergence. When the emergence disappears, all will not be observed, until the next rebirth, when the experiment can continue. Because of the discontinuous observation in time, we can never catch up with motion that is not moving toward us, so 5 directions on the spatial light cone will not be observed. In time, the emergence during the 2/3 period of our annihilation will also not be observed, and our observation results are forever limited to 1/18. This is also very easy to understand macroscopically, just as a person cannot see their own death, except that a person cannot immediately be reborn and cannot get this feeling. Particles, however, can quickly be reborn and re-emerge as slightly changed versions of us, which we perceive as the arrow of time.

3. Compatibility Statement

This theory does not conflict with existing theories, but only makes slight modifications. Existing theories and formulas remain applicable; for the time being, no further formulas will be deduced. Theory name: Void Phase Dynamics VPD (early on PVD was used), Void Phase Dynamics (VPD), early: The Phase Dynamics of the Void (PVD). Benchmarked against The Mathematical Principles of Natural Philosophy, The Physical Principles of Classical Philosophy.

4. Conclusion and Outlook

Void Phase Dynamics is a framework conjecture that runs through life and death, frontier science and technology, and classical philosophy, unifying the underlying layer of the universe. It describes us as living in a parallel universe of real-time annihilation and rebirth, with the universe divided into 6 directions in space and 3 phases in time. All spacetime (particles) are instantaneously annihilated and reborn; the only difference is the phase of rebirth. What determines the diversity of all things is the void and phase, which then emerge hierarchically: the first layer, void fluctuation, emerges as atoms; the second layer, atoms, emerges as molecular structures; the third layer emerges as consciousness. Void fluctuation $\rightarrow$ atom $\rightarrow$ structure $\rightarrow$ consciousness. Each layer forms a self-sustaining structure, and at the same time performs reverse transmission (gravity) to the underlying layer. This process is very similar to current artificial intelligence generation algorithms, with forward transmission (emergence) and reverse transmission (gravity). We are a huge self-generating computer.

To be continued.

References

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